

Simple & Easy AWS Core Concepts Notes

Yeh notes sirf aap ke bataye gaye specific topics par hain, taake aap easily revise kar sakain.

1. VPC (Virtual Private Cloud)

- **Easyst Explanation:** Yeh AWS ke andar aapka apna private, isolated network (ya bina kisi interfere ke aapka apna ilaka) hota hai.
- **Kaam:** Aapke saare servers (EC2), databases, aur storage isi VPC ke andar safe rehte hain.

2. Public IP vs Private IP

Feature	Public IP Address	Private IP Address
Kahan use hota hai?	Internet par (Outside world).	VPC ke andar (Internal network).
Access:	Internet se koi bhi connect kar sakta hai (agar SG allow kare).	Sirf VPC ke andar wale servers aapas mein baat kar sakte hain.
Changeability:	Server Stop/Start karne se badal jata hai (Dynamic).	Kabhi nahi badalta , hamesha same rehta hai.

3. EBS (Elastic Block Store)

- **Easyst Explanation:** Yeh EC2 Instance ki **Virtual Hard Disk (SSD/HDD)** hai.
- **Main Point:** Agar aapka EC2 server Stop bhi ho jaye, toh EBS drive ka data bilkul **safe** rehta hai.

4. Security Group (SG)

- **Easyst Explanation:** Yeh aapke EC2 Server ke darwaze par khada **Instance-Level Guard** hai.
- **Main Rules:**
 - Ismein sirf **ALLOW** rules hote hain (Block nahi kar sakta).
 - Check karta hai ke kis Port (e.g., Port 22 for SSH, Port 80 for Web) se traffic andar aaye.
 - Yeh **Stateful** hota hai (agar traffic andar aane diya, toh wapsi ka rasta khud hi khula)

rehta hai).

5. Key Pair (Chabi)

- **Easyst Explanation:** Server ke Operating System (OS/Linux) ke andar login karne ki **Chabi** (.pem ya .ppk file).
- **Work Flow:**
 - **Public Key:** AWS server par pehle se Lock ban kar set rehti hai.
 - **Private Key:** Aapke computer mein file hoti hai.
 - Lock aur Key match hote hain, tabhi aap Terminal par login kar sakti hain.

6. EC2 Lifecycle States

1. **Pending:** Server start hone ki tayyari kar raha hai.
2. **Running:** Server active hai, kaam chal raha hai. (**Billing start**).
3. **Stopping / Stopped:** Server Off hai (Jaise laptop Shut Down karna). **Compute bill 0 ho jata hai**, data EBS par safe rehta hai.
4. **Terminated:** Server permanent **Delete** ho chuka hai (Dubara wapas nahi aa sakta).

7. Pricing Models: On-Demand vs Reserved

- **On-Demand:**
 - **Rule:** "Jitna use karo, utna pay karo" (Pay-as-you-go).
 - **Best For:** Short-term testing, learning, aur flexible usage.
- **Reserved Instances (RI):**
 - **Rule:** 1 saal ya 3 saal ka pakka contract karna.
 - **Discount:** Continuous use ke badle AWS aapko **40% se 72% TAK SASTA (Discounted)** rate deta hai.

8. Scaling: Vertical vs Horizontal

- **Vertical Scaling (Scale Up / Down):**
 - Ek hi EC2 server ka size/power barhana (RAM aur CPU zyada karna, e.g. t2.micro → t2.large).
 - **Disadvantage:** Hardware change karne ke liye server **Stop** karna padta hai (Downtime).
- **Horizontal Scaling (Scale Out / In):**
 - Server ka size bada karne ke bajaye **zyada extra servers add kar dena** (Auto Scaling).
 - **Advantage:** **Zero Downtime**, capacity unlimited ho sakti hai.

9. High Availability (HA) in EC2

- **Easyst Explanation:** System ka aisa setup jo kabhi crash ya down na ho, 24/7 chalata rahe.

- **Kaise achieve hota hai?**
 1. Servers ko **Multiple Availability Zones (AZs / Data Centers)** mein rakhna.
 2. Front par **Load Balancer** aur **Auto Scaling** lagana taake agar ek Data Center down ho, toh doosra auto traffic sambhal le.

10. Shared Responsibility Model (AWS Manage vs You Manage)

AWS Manage = Security OF the Cloud | You Manage = Security IN the Cloud

- - **AWS Manage Karta Hai:**
 - Physical Data Center ki Security, Electricity, Cooling.
 - Physical Hardware (Physical Servers, Hard Drives, Cables).
 - Virtualization / Hypervisor layer.
 - **Aap (You) Manage Karte Hain:**
 - Operating System (Linux/Windows) ke security updates aur patches.
 - Firewall / Security Group aur Ports ki rules configuration.
 - Aapka Data, Files, aur Users ke Passwords (IAM).